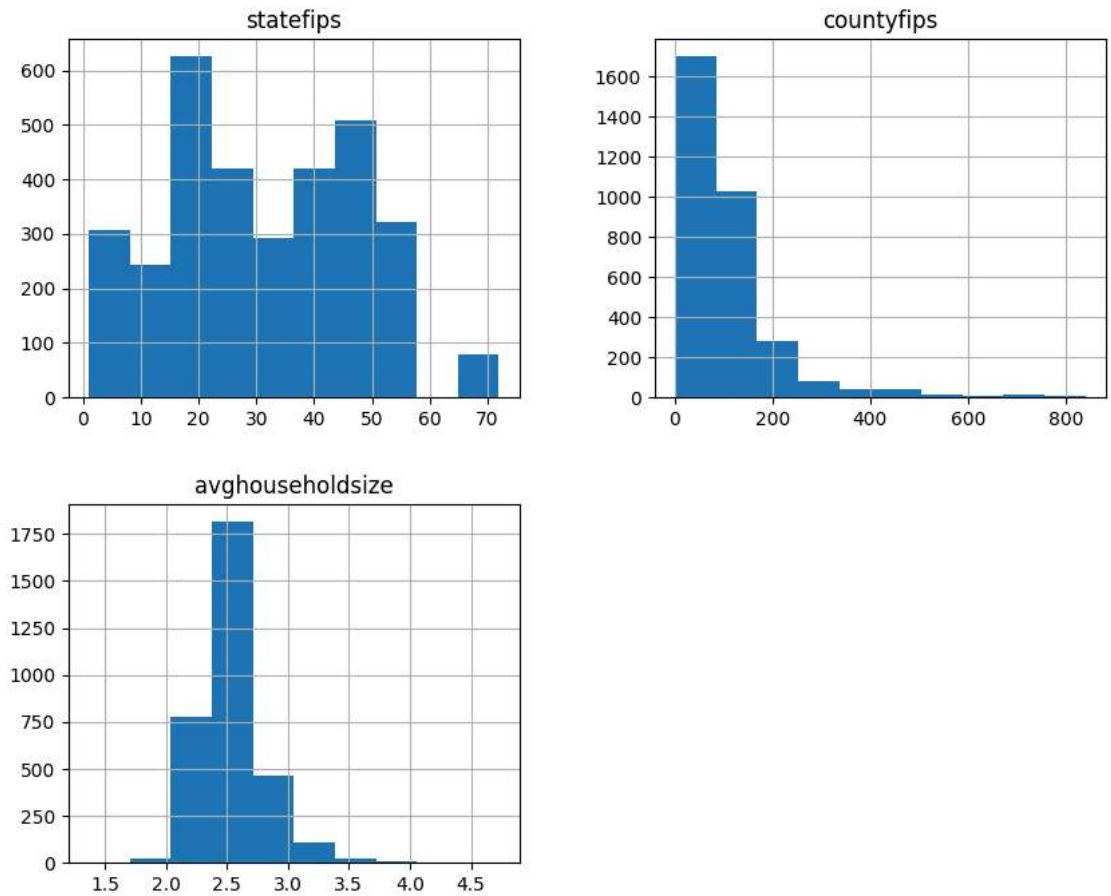
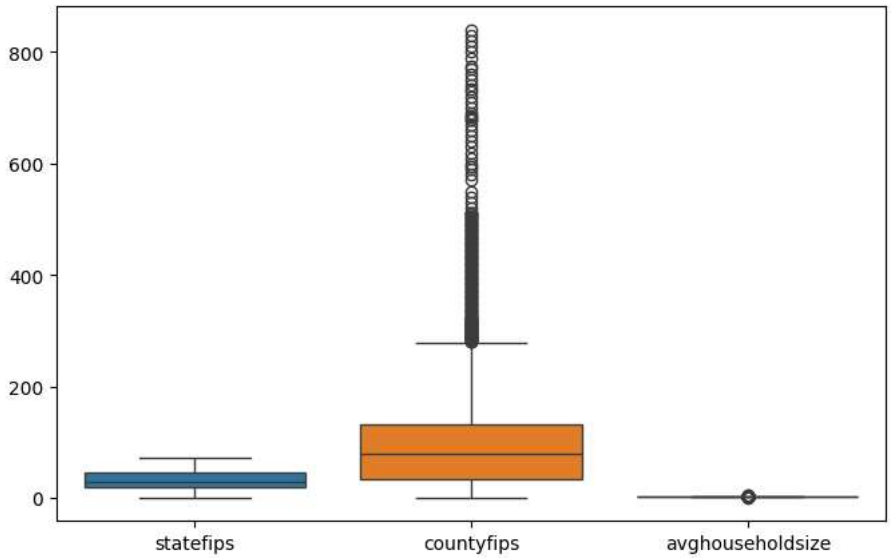
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.ensemble import RandomForestClassifier


df = pd.read_csv("avg-household-size.csv")


print(df.head())
print(df.info())
print(df.describe())
```

```
   statefips  countyfips  avghouseholdsize                           geography
0          2          13              2.43      Aleutians East Borough, Alaska
1          2          16              3.59  Aleutians West Census Area, Alaska
2          2          20              2.77      Anchorage Municipality, Alaska
3          2          50              3.86          Bethel Census Area, Alaska
4          2          60              2.50         Bristol Bay Borough, Alaska
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3220 entries, 0 to 3219
Data columns (total 4 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   statefips         3220 non-null   int64
 1   countyfips        3220 non-null   int64
 2   avghouseholdsize  3220 non-null   float64
 3   geography         3220 non-null   object
dtypes: float64(1), int64(2), object(1)
memory usage: 100.8+ KB
None
         statefips   countyfips  avghouseholdsize
count  3220.000000  3220.000000       3220.000000
mean     31.290683   102.922050          2.540755
std      16.277207   106.693846          0.267654
min       1.000000     1.000000          1.370000
25%      19.000000    35.000000          2.380000
50%      30.000000    79.000000          2.510000
75%      46.000000   133.000000          2.650000
max      72.000000   840.000000          4.730000
```

```
df.hist(figsize=(10,8))
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.boxplot(data=df.select_dtypes(include=np.number))
plt.show()
```




```
print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
   statefips  countyfips  avghouseholdsize                          geography
0          2          13              2.43      Aleutians East Borough, Alaska
1          2          16              3.59  Aleutians West Census Area, Alaska
2          2          20              2.77      Anchorage Municipality, Alaska
3          2          50              3.86          Bethel Census Area, Alaska
4          2          60              2.50         Bristol Bay Borough, Alaska
```

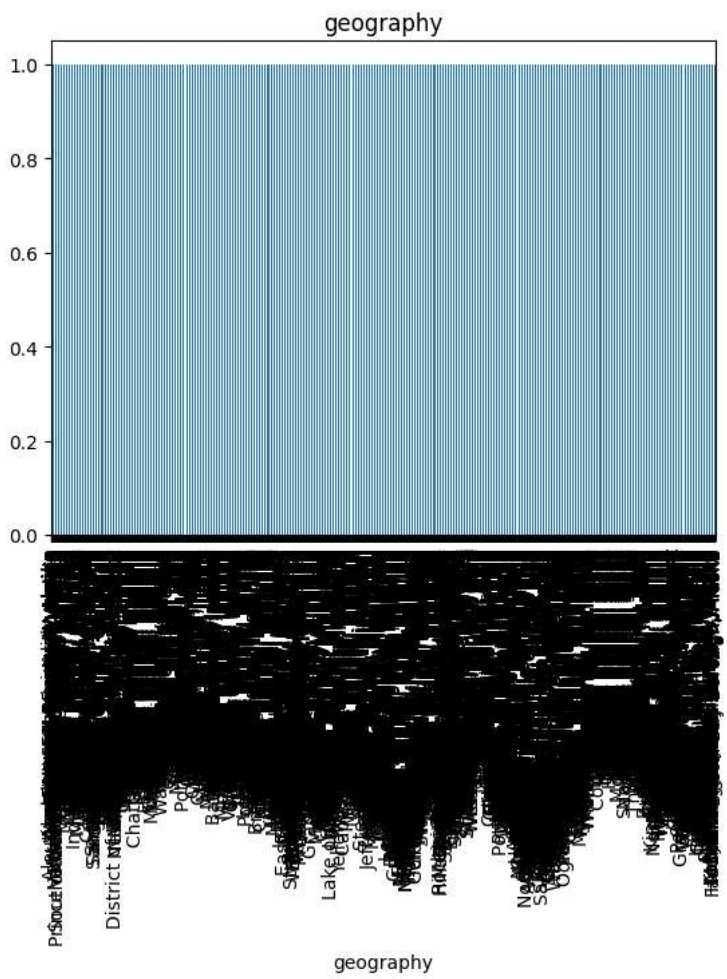
```
print("\nColumns:")
print(df.columns)
```

```
Columns:
Index(['statefips', 'countyfips', 'avghouseholdsize', 'geography'], dtype='object')
```

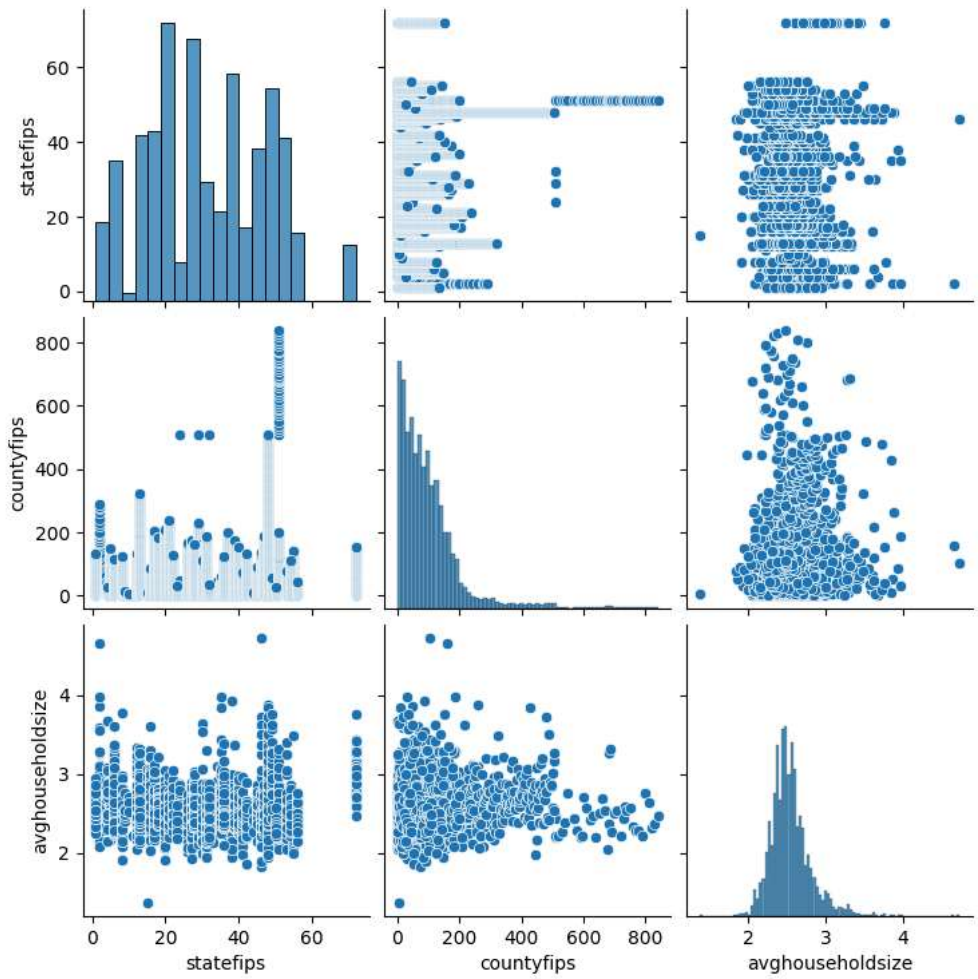
```
print("\nInfo:")
print(df.info())
```

```
Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3220 entries, 0 to 3219
Data columns (total 4 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   statefips         3220 non-null   int64
 1   countyfips        3220 non-null   int64
 2   avghouseholdsize  3220 non-null   float64
 3   geography         3220 non-null   object
dtypes: float64(1), int64(2), object(1)
memory usage: 100.8+ KB
None
```

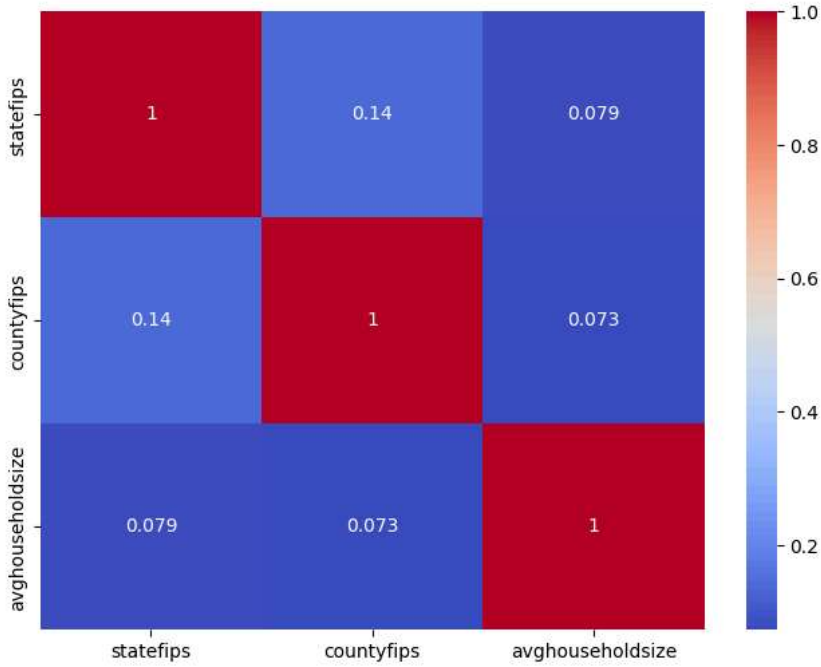
```
for col in df.select_dtypes(include='object').columns:
    df[col].value_counts().plot(kind='bar')
    plt.title(col)
    plt.show()
```




```
sns.pairplot(df)
plt.show()
```

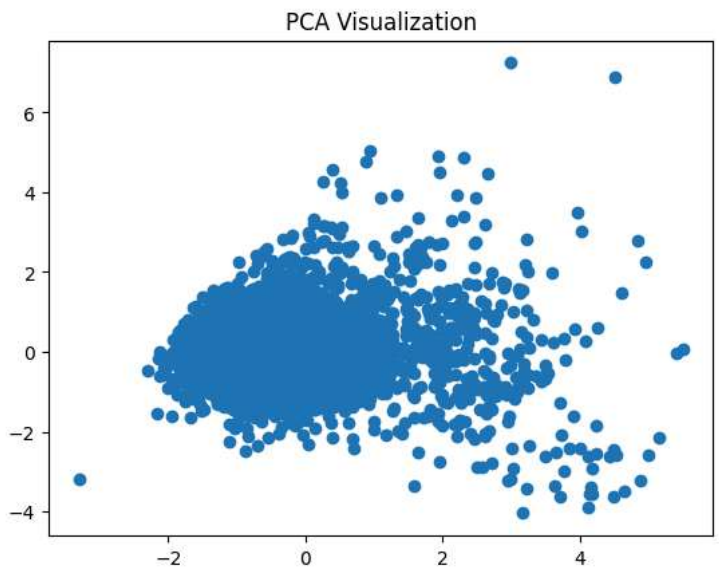



```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.show()
```




```
if len(df.select_dtypes(include='object').columns) >= 2:
    cols = df.select_dtypes(include='object').columns
    print(pd.crosstab(df[cols[0]], df[cols[1]]))
```

```
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

num_df = df.select_dtypes(include=np.number).dropna()

scaler = StandardScaler()
scaled = scaler.fit_transform(num_df)

pca = PCA(n_components=2)
pca_result = pca.fit_transform(scaled)

plt.scatter(pca_result[:,0], pca_result[:,1])
plt.title("PCA Visualization")
plt.show()
```




```
print(df.isnull().sum())


df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
statefips           0
countyfips          0
avghouseholdsize    0
geography           0
dtype: int64
```

```
le = LabelEncoder()
for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])
```

```
print("Mean:\n", df.mean())
print("Median:\n", df.median())
print("Std Dev:\n", df.std())
```

```
Mean:
 statefips             31.290683
countyfips           102.922050
avghouseholdsize       2.540755
geography           1609.500000
dtype: float64
Median:
 statefips             30.00
countyfips            79.00
avghouseholdsize       2.51
geography           1609.50
dtype: float64
Std Dev:
 statefips            16.277207
countyfips          106.693846
avghouseholdsize      0.267654
```

```
geography           929.678260
dtype: float64
```

```
cat_cols = df.select_dtypes(include='object').columns

if len(cat_cols) > 0:
    df[cat_cols[0]].value_counts().plot.pie(autopct='%1.1f%%')
    plt.title(cat_cols[0])
    plt.show()
```

```
from sklearn.metrics import r2_score, mean_squared_error
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()
model.fit(X_train, y_train)
```

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